network that extends to very large distance to connect the computers. One example of WAN is Internet. These cables are placed in deep underground to avoid any damage to the cables. The optic cable uses light to transmit the information from one place to another. There are two types of fiber optic cables available, One is single-mode (100BaseBx) another one is Multimode (100BaseSX). Single-mode cables are used for long distance transmission and at a high cost whereas the multimode cables are used for short distance transmission at a very low cost. The optic cables are easy to maintain and install. See Figure 13.4



Figure 13.4 Fiber optic cable with 4 cores combined with one cable.

4. **USB Cables:** The Universal Serial Bus are used to connect keyboard, mouse and other peripheral devices. But there are some special network devices used to connect the Internet through the USB called dongles. The dongle is a small peripheral device which has a compatible of mobile broadband with a slim slot in it and connects the Internet and acts as a modem to the computer. See Figure 13.5

DO YOU KNOW?

The latest version of USB is USB 3.0 which has the data transfer rate 4.85 Gbps. But USB 2.0 has just 480 Mbps.

Micro USB is a miniaturized version of the USB used for connecting mobile devices such as smart phones, GPS devices and digital cameras.



5. **Serial and Parallel cables:** Earlier in 1980s to 1990s the Ethernet and the USB were not developed. Then the Serial and Parallel interface cables were used to connect the Internet to the system. They were sometime used for PC to PC networking. Before the USB emerged, the system will have both serial port and parallel port. The serial port will send 1 bit at one time whereas the parallel port will send 8 bit at one time. The parallel cables are used to connect to



Figure 13.5 The USB Cables and USB dongle connected to a laptop.

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